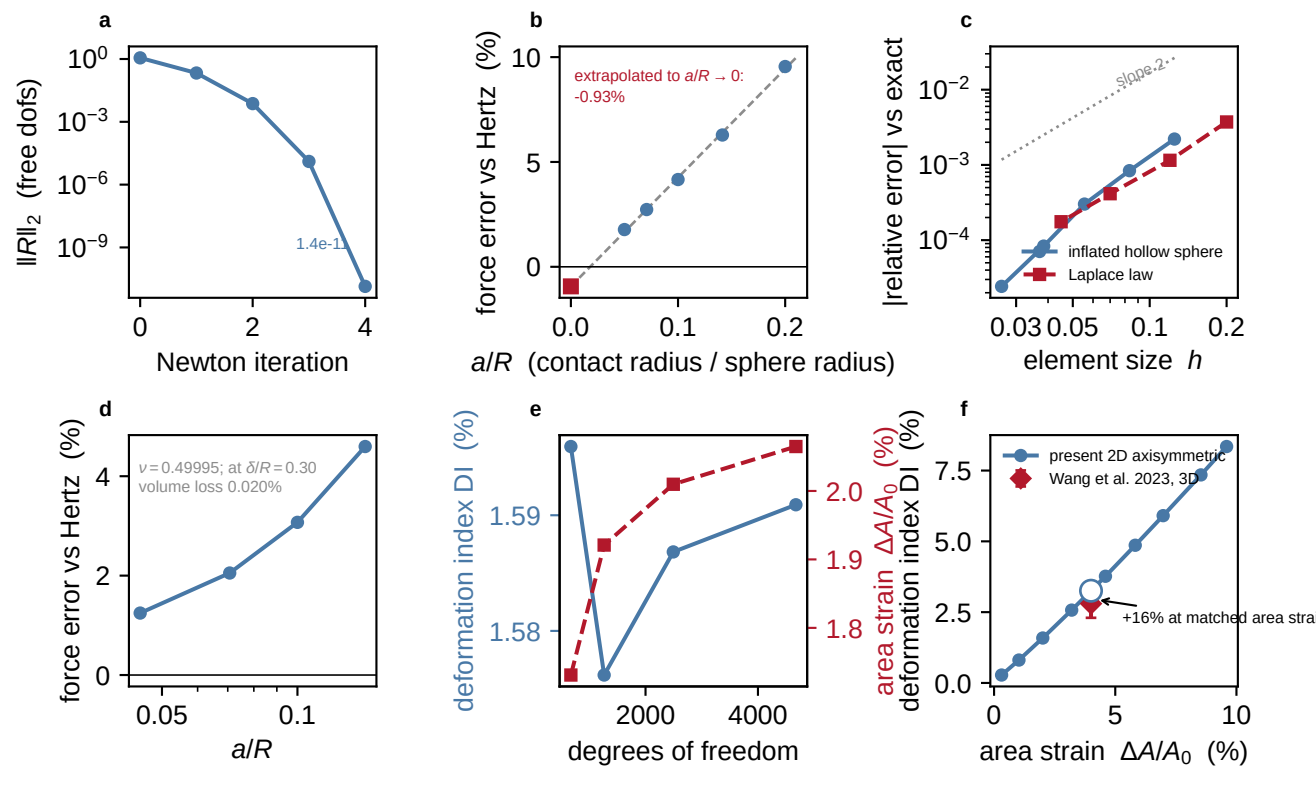
a
$\|R\|_2$ (free dofs)
10^0
10^{-3}
10^{-6}
10^{-9}
1.4e-11
0
2
4
Newton iteration
b
force error vs Hertz (%)
extrapolated to $a/R \to 0$:
-0.93%
10
5
0
0.0
0.1
0.2
a/R (contact radius / sphere radius)
c
|relative error| vs exact
slope 2
10^{-2}
10^{-3}
10^{-4}
inflated hollow sphere
Laplace law
0.03
0.05
0.1
0.2
element size h
d
force error vs Hertz (%)
$\nu = 0.49995$; at $\delta/R = 0.30$
volume loss 0.020%
4
2
0
0.05
0.1
a/R
e
deformation index DI (%)
1.59
1.58
area strain $\Delta A/A_0$ (%)
2.0
1.9
1.8
2000
4000
degrees of freedom
f
deformation index DI (%)
present 2D axisymmetric
Wang et al. 2023, 3D
+16% at matched area stra
7.5
5.0
2.5
0.0
0
5
10
area strain $\Delta A/A_0$ (%)